

Purunjay Pratap Singh

AI Engineer

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tinyurl.com/2bpen54v | Bhopal, MP

EDUCATION

Vellore Institute of Technology

B.Tech in Computer Science & Engineering (AI & Machine Learning)

2023 – 2027

Bhopal, MP

SKILLS

Programming : Python (primary), JavaScript (secondary)

Databases : SQL, PostgreSQL, Vector Search

Frameworks : FastAPI, asyncio, WebSockets, Node.js, PyTorch, scikit-learn, SetFit, Sentence-Transformers, OpenCV, Streamlit

Concepts : Machine Learning, NLP, RAG, LLM Tool Calling, Voice AI (STT/TTS), Computer Vision

Tools : Docker, Git

INTERNSHIP

AI/ML Intern, Sify Technologies Limited

Jun. 2026 – Aug. 2026 (2 months)

Applied NLP/LLM engineering for the enterprise USP platform

Chennai, TN (On-site)

- Built an address matching engine (Python, rapidfuzz, FastAPI, PostgreSQL) that reached **96% accuracy on 3,678 query variants at sub-100 ms** latency, and cut CPU use **38%** by profiling the scoring loop.
- Rebuilt the internal RAG support chatbot on BM25 retrieval over 13 documentation guides, cutting context from 15k to **~4k tokens per query (70% reduction)**, and hardened it against prompt injection and XSS with automated tests.
- Shipped a vision-LLM invoice/PO extraction service (FastAPI, PyMuPDF, Tesseract) that turns scanned or digital PDFs into a strict **9-field line-item schema**, with retries, model rotation, and truncation guards that fail loudly instead of fabricating data.

PROJECTS

Real-Time Bilingual Voice Agent Telephony | Python, asyncio, WebSockets, LLM + RAG | <https://tinyurl.com/3fecvz9p>

- Built and run a production outbound phone agent for a private client that places live PSTN calls (8 kHz μ -law media WebSocket), follows callers from English into Hindi mid-sentence, answers from a vector-search knowledge base, and books meetings via LLM tool calls, logging every call to a built-in CRM.
- Profiled the STT $\rightarrow$ LLM $\rightarrow$ TTS cascade at a **~2.19 s** round trip and showed the LLM (0.67 s median TTFT) was not the bottleneck; first-token TTS streaming saved **200–300 ms per sentence**, and rolling context summarization stopped quadratic prompt growth (7,278 tokens by turn three).
- Hardened it for live lines: single-owner turn detection to end STT/VAD conflicts, stepped LLM failure recovery with clean hang-up, a streaming filter that keeps tool-call markup from being read aloud, and headless evals that run the real pipeline without placing calls.

Invoice / PO Extraction Service | Python, Vision LLM, FastAPI, Tesseract | tinyurl.com/3h2vdu43

- Built a vision-LLM document extraction service converting invoice and PO PDFs, digital or scanned, into a fixed 9-field line-item CSV with no templates and no per-vendor rules.
- Added reliability guards around unstable endpoints: model rotation on rate limits, truncation and decoding-loop detection, and an arithmetic integrity check on totals.
- Cut inference cost with a keep-biased page filter classifying out non-billing pages via local OCR before they become image tokens; validated on real invoices from three countries.

Mandi: Multilingual Voice Shopping Assistant | Python, FastAPI, Pydantic, Next.js | tinyurl.com/3bjm4ftw

- Built a voice shopping list handling English, Hindi, and code-switched Hinglish (“do kilo aloo add karo” $\rightarrow$ 2 kg potatoes, categorised), where every utterance collapses into one schema-constrained Pydantic intent object that satisfies 8 required features from a single design.
- Routed 86% of commands through a deterministic on-device parser at 0 ms and zero LLM quota, deferring to a ranked free-model chain only for open-ended speech; each answer is labelled on-device, model, or offline, and the app stays usable with no API key.
- Reached 99.6% field-level accuracy on a 56-case eval suite (100% on the 220-field deterministic subset) backed by a resilience ladder: strict JSON schema first, plain-JSON retry, one repair pass on validation failure, and rate-limited models skipped rather than retried.

CERTIFICATIONS

Google Cloud Gen AI Exchange Program, Hack2skill, Jun. 2025; Credential ID 2025H2S04GENAI-A100248: Vertex AI, Gemini APIs.

NPTEL Elite: Cloud Computing, IIT Kharagpur, MoE Govt. of India, May 2025; Credential ID NPTEL25CS11S942800871: 25/25 on all assignments; top performer among 29,703 certified candidates.

Google IT Support Certificate, Google Cloud Skills Boost, Feb. 2026; tinyurl.com/eyezpnuv: networking, system administration, IT security, operating systems, troubleshooting.